

Exp7: Deploy a Simple Application on a Local Kubernetes Cluster using Minikube.

Aim

To deploy a simple application on a **local Kubernetes cluster** using **Minikube**.

Software Requirements

- Ubuntu 22.04 LTS
- Docker (already installed)
- Minikube
- kubectl

PART A: INSTALL & START MINIKUBE

Step 1: Install kubectl

Create a project directory

```
mkdir app-in-minikube
```

```
cd app-in-minikube
```

```
cse@cse-B760M-H:~$ mkdir app-in-minikube
cse@cse-B760M-H:~$ cd app-in-minikube
cse@cse-B760M-H:~/app-in-minikube$
```

Install curl

```
sudo snap install curl
```

```
cse@cse-B760M-H:~/app-in-minikube$ sudo snap install curl
[sudo] password for cse:
curl 8.18.0 from Yuzukosho (aoilinux) installed
cse@cse-B760M-H:~/app-in-minikube$
```

```
curl --version
```

```
cse@cse-B760M-H:~/app-in-minikube$ curl --version

Caution: You are using the Snap version of curl.
Due to Snap's sandbox nature, this version has some limitations.
For example, it may not be able to access hidden folders in your home directory
or other restricted areas of the OS.

Which means you may encounter errors when using snap curl to download and execute some script.
For those cases, you might want to use the native curl package.
For details, see: https://github.com/boukendesho/curl-snap/issues/1

To stop seeing this message, run the following command:
$ curl.snap-ack

curl 8.18.0 (x86_64-pc-linux-gnu) libcurl/8.18.0 OpenSSL/3.6.0 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.21.2 libssh2/1.11.0 nghttp2/1.68.0 nghttp3/1.15.0 librtmp/2.3 mlt-krb5/1.20.1 OpenLDAP/2.6.7
Release-Date: 2026-01-07
Protocols: dict file ftp ftps gopher gophers http https imap imaps lpfs lpns ldap ldaps mqtt pop3 pop3s rtsp rtsp scp sftp smb snbs snmp snmps telnet tftp ws wss
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTP3 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
```

```
curl -LO https://storage.googleapis.com/kubernetes-  
release/release/$(curl -s  
https://storage.googleapis.com/kubernetes-  
release/release/stable.txt)/bin/linux/amd64/kubectl
```

```
cse@cse-B760M-H:~/app-in-minikube$ curl -LO https://storage.googleapis.com/kubernetes-release/release/$(curl -s https://storage.googleapis.com/kubernetes-release/release/stable.txt)/bin/linux/amd64/kubectl

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For those cases, you might want to use the native curl package.
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To stop seeing this message, run the following command:
$ curl.snap-acked

% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 53.76M 100 53.76M    0     0  0 19.70M      0  0:04  0:04  0:04  9.11M
cse@cse-B760M-H:~/app-in-minikube$
```

```
ls -la
```

```
cse@cse-B760M-H:~/app-in-minikube$ ls -la
total 55072
drwxrwxr-x  2 cse cse      4096 Feb 20 16:12 .
drwxr-x--- 27 cse cse      4096 Feb 20 16:03 ..
-rwxrwxr-x  1 cse cse 56381592 Feb 20 16:12 kubectl
cse@cse-B760M-H:~/app-in-minikube$
```

```
chmod +x kubectl
```

```
cse@cse-B760M-H:~/app-in-minikube$ chmod +x kubectl
cse@cse-B760M-H:~/app-in-minikube$
```

```
sudo mv kubectl /usr/local/bin/
```

```
cse@cse-B760M-H:~/app-in-minikube$ sudo mv kubectl /usr/local/bin/
cse@cse-B760M-H:~/app-in-minikube$
```

Verify:

```
kubectl version --client
```

```
cse@cse-B760M-H:~/app-in-minikube$ kubectl version --client
Client Version: v1.31.0
Kustomize Version: v5.4.2
cse@cse-B760M-H:~/app-in-minikube$
```

Step 2: Install Minikube

```
curl -LO  
https://storage.googleapis.com/minikube/releases/latest/minikube-  
linux-amd64
```

```
cse@cse-B760M-H:~/app-in-minikube$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64

Caution: You are using the Snap version of curl.
Due to Snap's sandbox nature, this version has some limitations.
For example, it may not be able to access hidden folders in your home directory
or other restricted areas of the os.

Which means you may encounter errors when using snap curl to download and execute some script.
For those cases, you might want to use the native curl package.
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To stop seeing this message, run the following command:
$ curl.snap-acked

  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 128.6M 100 128.6M    0     0  14.71M    0     0    0:08    0:08    0:08 18.47M
cse@cse-B760M-H:~/app-in-minikube$
```

```
ls -la
```

```
cse@cse-B760M-H:~/app-in-minikube$ ls -la
total 131744
drwxrwxr-x  2 cse cse      4096 Feb 20 16:17 .
drwxr-x--- 27 cse cse      4096 Feb 20 16:03 ..
-rw-rw-r--  1 cse cse 134891912 Feb 20 16:17 minikube-linux-amd64
cse@cse-B760M-H:~/app-in-minikube$
```

```
chmod +x minikube-linux-amd64
```

```
ls -la
```

```
cse@cse-B760M-H:~/app-in-minikube$ chmod +x minikube-linux-amd64
cse@cse-B760M-H:~/app-in-minikube$
cse@cse-B760M-H:~/app-in-minikube$
cse@cse-B760M-H:~/app-in-minikube$
cse@cse-B760M-H:~/app-in-minikube$
cse@cse-B760M-H:~/app-in-minikube$ ls -la
total 131744
drwxrwxr-x  2 cse cse      4096 Feb 20 16:17 .
drwxr-x--- 27 cse cse      4096 Feb 20 16:03 ..
-rwxrwxr-x  1 cse cse 134891912 Feb 20 16:17 minikube-linux-amd64
cse@cse-B760M-H:~/app-in-minikube$
```

```
sudo mv minikube-linux-amd64 /usr/local/bin/minikube
```

```
cse@cse-B760M-H:~/app-in-minikube$ sudo mv minikube-linux-amd64 /usr/local/bin/minikube
cse@cse-B760M-H:~/app-in-minikube$
```

Verify:

minikube version

```
cse@cse-B760M-H:~/app-in-minikube$ minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
cse@cse-B760M-H:~/app-in-minikube$
```

Step 3: Start Minikube Cluster

```
minikube start --driver=docker
```

Note: It takes 2-5 minutes for first time start.

```
cse@cse-B760M-H:~/app-in-minikube$ minikube start --driver=docker
🐳 minikube v1.38.1 on Ubuntu 22.04
🌟 Using the docker driver based on user configuration
👉 Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
👉 Using Docker driver with root privileges
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.50 ...
📦 Downloading Kubernetes v1.35.1 preload ...
> gcr.io/k8s-minikube/kicbase...: 519.56 MiB / 519.58 MiB   100.00% 53.34 M
> preloaded-images-k8s-v18-v1...: 272.45 MiB / 272.45 MiB   100.00% 17.21 M
🔥 Creating docker container (CPUs=2, Memory=3900MB) ...
🔧 Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔧 Configuring bridge CNI (Container Networking Interface) ...
🔧 Verifying Kubernetes components...
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   🌟 Enabled addons: storage-provisioner, default-storageclass

! /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.35.1.
   ■ Want kubectl v1.35.1? Try 'minikube kubectl -- get pods -A'
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
cse@cse-B760M-H:~/app-in-minikube$
```

Step 4: Verify Cluster Status

```
kubectl get nodes
```

```
cse@cse-B760M-H:~/app-in-minikube$ kubectl get nodes
NAME        STATUS    ROLES    AGE   VERSION
minikube    Ready     control-plane  55s   v1.35.1
cse@cse-B760M-H:~/app-in-minikube$
```

PART B: DEPLOY A SIMPLE APPLICATION

Let us deploy a **simple NGINX web application**.

Step 5: Create Deployment

```
kubectl create deployment web-app --image=nginx
```

```
cse@cse-B760M-H:~/app-in-minikube$ kubectl create deployment web-app --image=nginx
deployment.apps/web-app created
cse@cse-B760M-H:~/app-in-minikube$
```

Step 6: Verify Deployment

```
kubectl get deployments
```

```
cse@cse-B760M-H:~/app-in-minikube$ kubectl get deployments
NAME          READY    UP-TO-DATE    AVAILABLE    AGE
web-app       1/1      1             1            18s
cse@cse-B760M-H:~/app-in-minikube$
```

```
kubectl get pods
```

```
cse@cse-B760M-H:~/app-in-minikube$ kubectl get pods
NAME                                READY    STATUS    RESTARTS    AGE
web-app-5d4dd47c87-24t4d           1/1      Running   0           47s
cse@cse-B760M-H:~/app-in-minikube$
```

Step 7: Expose Deployment as a Service

```
kubectl expose deployment web-app --type=NodePort --port=80
```

```
cse@cse-B760M-H:~/app-in-minikube$ kubectl expose deployment web-app --type=NodePort --port=80
service/web-app exposed
cse@cse-B760M-H:~/app-in-minikube$
```

Step 8: Get Service Details

```
kubectl get services
```

```
cse@cse-B760M-H:~/app-in-minikube$ kubectl get services
NAME          TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes    ClusterIP     10.96.0.1     <none>         443/TCP          4m36s
web-app       NodePort      10.107.86.4   <none>         80:30295/TCP     27s
cse@cse-B760M-H:~/app-in-minikube$
```

Step 9: Access Application in Browser

```
minikube service web-app
```

Note:

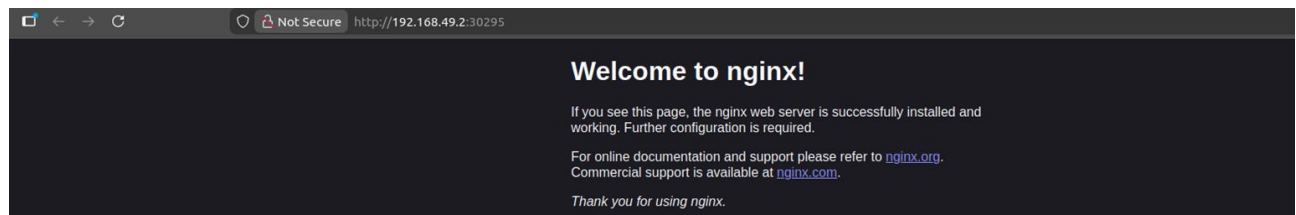
Browser opens automatically

NGINX welcome page visible

```
cse@cse-B760M-H: /app-in-minikube$ minikube service web-app
```

NAMESPACE	NAME	TARGET PORT	URL
default	web-app	80	http://192.168.49.2:30295

```
Opening service default/web-app in default browser...
cse@cse-B760M-H: /app-in-minikube$ Gtk+Message: 16:27:04.495: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
```



Access application using curl

Copy URL and use it with curl

```
curl http://192.168.49.2:30295
```

```
cse@cse-B760M-H: /app-in-minikube$ curl http://192.168.49.2:30295

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For example, it may not be able to access hidden folders in your home directory
or other restricted areas of the os.

Which means you may encounter errors when using snap curl to download and execute some script.
For those cases, you might want to use the native curl package.
For details, see: https://github.com/boukendescho/curl-snap/issues/1

To stop seeing this message, run the following command:
$ curl.snap-acked

<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
cse@cse-B760M-H: /app-in-minikube$
```

PART C: CLEANUP (GOOD PRACTICE)

Step 10: Delete Deployment & Service

```
kubectl delete service web-app
```

```
kubectl get services
```

```
cse@cse-B760M-H:~/app-in-minikube$ kubectl delete service web-app
service "web-app" deleted
cse@cse-B760M-H:~/app-in-minikube$ kubectl get services
NAME           TYPE          CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
kubernetes     ClusterIP     10.96.0.1     <none>       443/TCP    18m
cse@cse-B760M-H:~/app-in-minikube$
```

```
kubectl delete deployment web-app
```

```
kubectl get deployments
```

```
cse@cse-B760M-H:~/app-in-minikube$ kubectl delete deployment web-app
deployment.apps "web-app" deleted
cse@cse-B760M-H:~/app-in-minikube$ kubectl get deployments
No resources found in default namespace.
cse@cse-B760M-H:~/app-in-minikube$
```

```
kubectl get pods
```

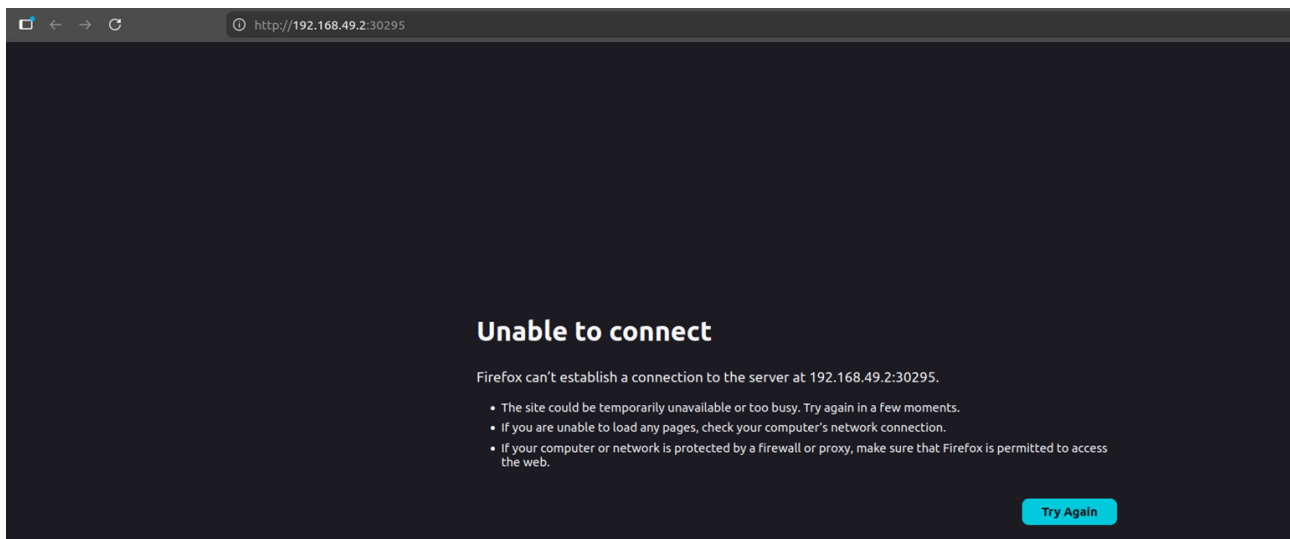
```
cse@cse-B760M-H:~/app-in-minikube$ kubectl get pods
No resources found in default namespace.
cse@cse-B760M-H:~/app-in-minikube$
```

Step 11: Stop Minikube

```
minikube stop
```

```
cse@cse-B760M-H:~/app-in-minikube$ minikube stop
👉 Stopping node "minikube" ...
🔴 Powering off "minikube" via SSH ...
🔴 1 node stopped.
cse@cse-B760M-H:~/app-in-minikube$
```

Go back to browser and refresh the page to make sure that application is not running.



Result

A simple application was successfully deployed on a local Kubernetes cluster using Minikube.